

Data Mining Project Ideas

Overview

This project aims to provide students with practical experience in data mining, web mining, and graph-based analysis techniques. Students are required to collect data from real-world sources, such as web scraping or publicly available datasets, and then apply data mining algorithms to extract meaningful patterns, relationships, and insights.

The project integrates three main concepts:

- Web Mining (Data Collection & Graph Construction)
- Association Rule Mining (Apriori / FP-Growth)
- Link Analysis (PageRank / HITS)

Students will also perform data visualization and interpret results to support decision-making. Additionally, students will integrate BERT-based deep learning techniques for text classification or similarity analysis to enhance the quality of insights.

1. E-commerce Recommendation System.

This project focuses on building an intelligent recommendation system using data collected either through web scraping or from publicly available datasets related to Amazon. The main idea is to collect product information, such as name, category, price, and customer reviews, and then construct a transaction dataset in which each record represents items purchased together by a user. This dataset will be used as the foundation for discovering hidden patterns.

Task 1: Data Mining & Graph Analysis

To extract useful knowledge, students will apply:

- Apriori or FP-Growth → to discover frequently bought product combinations
- PageRank algorithm → to rank products based on their importance

The results should be visualized using bar charts, association graphs, and network diagrams. The final output will include **a recommendation system suggesting products like “Customers who bought X also bought Y”**, along with insights about product combinations and high-demand categories.

Task 2: BERT-based Analysis (5)

Students will use BERT to analyze textual product data such as descriptions and customer reviews. This includes:

- Sentiment classification of reviews (positive, negative, neutral)

The final output will enhance the recommendation system by incorporating semantic understanding of products, allowing more accurate and personalized recommendations.

2. News Website Link Analysis

This project aims to analyze the structure of online news websites by collecting data through web scraping from platforms such as BBC or CNN. Students will crawl multiple articles and extract hyperlinks between them to build a web graph representing connections between pages.

Task 1: Link Analysis

Apply:

- PageRank algorithm → to identify the most important articles
- HITS algorithm → to distinguish between hub pages and authoritative pages

Visualization will include network graphs showing connections between articles and ranking charts. The final output will be a **ranked list of influential pages, supported by insights explaining how information flows and which pages drive traffic.**

Task 2: BERT-based Analysis

Students will apply BERT on article content to perform:

- Text classification (e.g., politics, sports, technology)

The final output will include categorized and clustered articles, providing deeper insight into content structure beyond hyperlinks.

3. Movie Recommendation System

This project focuses on analyzing user viewing behavior using datasets inspired by Netflix. Students will work with user watch histories to build a dataset where each record represents a list of watched movies.

Task 1: Pattern Mining

To discover patterns, students will apply:

- Apriori or FP-Growth → to find frequently watched movie combinations
- PageRank → to rank important or popular movies

Visualizations such as genre distribution charts and frequent movie pair graphs will be used to support the analysis.

Task 2: BERT-based Analysis

Students will use BERT on movie descriptions and summaries to perform **Similarity analysis between movies**.

The final system will recommend movies based on user behavior, along with insights about popular genres and viewing trends.

4. GitHub Repository Mining

This project involves analyzing software development trends by collecting repository data from GitHub. Students will extract information such as repository names, programming languages, stars, forks, and topics, then build relationships between repositories based on shared technologies.

Task 1: Graph & Pattern Analysis

The analysis will include:

- PageRank algorithm → to identify influential repositories
- Apriori or FP-Growth → to find commonly used technology combinations

Results will be visualized using charts for programming trends and network graphs for repository connections. **The final output will highlight popular technologies and suggest effective tech stacks, supported by insights into current development trends.**

Task 2: BERT-based Analysis

Students will apply BERT on repository descriptions and README files to perform **Text classification (e.g., AI, web development, mobile applications)**

The final output will enhance understanding of repositories and recommend similar projects or technologies.

5. Food Delivery Pattern Analysis

This project explores customer ordering behavior using data collected from platforms like Talabat or Uber Eats. Students will collect data about meals, categories, and orders, then construct a dataset where each transaction represents items ordered together.

Task 1: Transaction Mining

To analyze patterns, students will apply:

- Apriori or FP-Growth → to identify frequently ordered meal combinations
- PageRank → to rank popular meals

Task 2: BERT-based Analysis

Students will use BERT on meal descriptions and customer reviews to perform **Sentiment analysis of customer feedback.**

The final output will include a recommendation system for meal combos, along with insights into customer preferences and best-selling items.

6. Online Learning Behavior Analysis

This project analyzes student learning patterns using datasets from platforms such as Coursera. Students will examine course enrollment data to understand how learners progress through different courses.

Task 1: Pattern & Graph Analysis

The analysis will involve:

- Apriori or FP-Growth → to identify common course combinations
- PageRank algorithm → to determine important or influential courses

Task 2: BERT-based Analysis

Students will apply BERT on course descriptions to perform **similarity analysis between courses**

The final output will suggest recommended learning paths, supported by insights into high-demand skills and common course sequences.

7. Social Media Influencer Analysis

This project examines social networks by collecting data from platforms like Twitter or Instagram. Students will build a network of users based on followers and interactions, as well as extract hashtags from posts.

Task 1: Network Analysis

The analysis will involve:

- PageRank or HITS → to identify influential users
- Apriori or FP-Growth → to discover common hashtag combinations

Task 2: BERT-based Analysis

Students will use BERT on posts and tweets to perform **Sentiment analysis**

The final output will rank influencers and identify trending topics, supported by insights into user influence and content patterns.

Grading

Task 1

Data Collection & Understanding	2
Data Preprocessing	2
Association Rule Mining	2
Link Analysis (PageRank/HITS)	2
Visualization	1
Report & Presentation	1
Total Marks	10

Task 2

BERT Model	5 Marks
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